hea-bench: An open benchmark suite and reference baselines for high-entropy alloy phase prediction

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Abstract

Empirical phase-prediction rules for high-entropy alloys (HEAs) — the entropy criterion of Yeh, the atomic-size mismatch criterion of Zhang, the valence-electron-concentration criterion of Guo and Liu, and the Ω-parameter criterion of Yang and Zhang — are widely used as fast pre-screens for candidate compositions. Their quoted accuracies, however, come from heterogeneous, mutually incompatible datasets, and no published artifact treats them as proper diagnostic classifiers on a common open benchmark. We present `hea-bench`, a Python library and reproducible benchmark suite that (i) consolidates three primary open HEA-phase datasets (Borg et al. 2020; Pei et al. 2020; Peivaste et al. 2023) into a single 7784-composition resource with explicit per-row source provenance and cross-source label-conflict tracking, (ii) implements all four canonical empirical rules as binary or multi-class diagnostic classifiers with full diagnostic statistics (accuracy, sensitivity, specificity, Wilson 95% intervals, receiver-operating-characteristic curves, Youden's J), and (iii) ships in two interchangeable user surfaces: a standard `pip`-installable Python package and an in-browser frontend that runs the same wheel via Pyodide / WebAssembly, requiring no local install. On the consolidated benchmark, both binary rules collapse to "predict single-phase almost always" (Youden's J of 0.075 and 0.032 respectively), and a receiver-operating-characteristic sweep of the Zhang δ threshold finds an optimum at $\delta < 2.5\%$ rather than the canonical $\delta < 6.5\%$. We release `hea-bench` 0.0.1 with 151 tests, including pinned regression tests on every headline benchmark number, under the MIT License.

Keywords: high-entropy alloys, multi-principal element alloys, phase prediction, benchmark suite, empirical descriptors, Miedema model, Pyodide, open data, diagnostic classifiers

Required Metadata

1. Motivation and significance

High-entropy alloys (HEAs), introduced independently by Cantor et al. [1] and Yeh et al. [2] in 2004, are multi-component alloys (typically five or more principal elements near equiatomic composition) in which the configurational mixing entropy is large enough to stabilize disordered solid solutions over the intermetallic compounds predicted by classical Hume-Rothery rules. The compositional space is vast — combinatorially larger than that of conventional alloys — which makes *pre-screening* a practical necessity before any expensive density-functional-theory, CALPHAD, or experimental campaign.

Four empirical phase-prediction rules dominate the literature: the configurational mixing entropy criterion of Yeh ($\Delta S_{\text{mix}} > 1.5R$ [2]); the atomic-size mismatch criterion of Zhang ($\delta < 6.5\%$ [3]); the valence-electron concentration criterion of Guo and Liu (FCC favored at VEC ≥ 8.0, BCC at VEC < 6.87 [4]); and the Ω-parameter criterion of Yang and Zhang ($\Omega = T_{\text{m}} \cdot \Delta S_{\text{mix}}/|\Delta H_{\text{mix}}| > 1.1$ [5]). Each is supported in its original publication by classification rates of roughly 80% on the dataset available at the time.

Three gaps in the open literature motivate this work. **First,** every published evaluation of these rules uses a different dataset, with different preprocessing, dedup, and phase labeling. Head-to-head comparison is therefore impossible. To our knowledge there is no published, open, MatBench-style benchmark suite [6] for HEA phase classification. **Second,** the rules are typically reported as "accuracies" of the form X/N, without the diagnostic-classifier machinery (sensitivity, specificity, Wilson intervals, ROC analysis, Youden's J) that would expose where they actually generalize. **Third,** the only field-standard interactive tool for Miedema-style enthalpy calculations is the Windows-only, registration-gated MiedCalc [7]; no zero-install open analogue exists.

`hea-bench` addresses all three. We consolidate three primary open HEA-phase datasets into a single 7784-composition resource with full provenance, implement the four canonical rules as proper diagnostic classifiers, and ship the result in two equivalent surfaces: a standard Python library and an in-browser frontend that runs the same wheel via Pyodide.

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Table 1: Code metadata.

Nr.	Code metadata description
C1 Current code version	0.0.1
C2 Permanent link to code/repository	to be assigned via Zenodo on first tagged release
C3 Permanent link to reproducible capsule	N/A (the package is dependency-free in its core; reproducible via `pip install hea-bench`)
C4 Legal code license	MIT
C5 Code versioning system used	git
C6 Software code languages, tools, and services used	Python ($\geq$3.10), Pyodide (browser deployment)
C7 Compilation requirements, operating environments & dependencies	Pure Python, dependency-free core; optional extras: `pandas`, `matplotlib`, `pytest`
C8 If available, link to developer documentation/manual	`README.md` in the repository
C9 Support email for questions	`dfieser9@gmail.com`

2. Software description

2.1. Software architecture

`hea-bench` is organized into four orthogonal sub-packages plus a thin command-line interface:

- `hea_bench.composition` — formula parsing and composition normalization. A single regular-expression-based parser handles all three source formula conventions (Borg's space-separated proportional amounts, Pei's compact form, Peivaste's bare element strings).

- `hea_bench.descriptors` — the six empirical descriptors used by the rule set: ΔS_{mix}, δ, VEC, T_{m}, ΔH_{mix} (Miedema, multi-component [8, 9]), and Ω [5]. Per-element parameters live in `descriptors/data/elemental.py` (24 elements); binary mixing-enthalpy pairs are loaded from a vendored copy of matminer's [10] Miedema pair table (BSD-3-Clause, license preserved alongside the data file), which itself derives from the de Boer parameter set [11] and is consistent with Takeuchi and Inoue's open tabulation [12].

- `hea_bench.rules` — the four canonical empirical rules wrapped as classifiers, each in its own module (`yeh_smix`, `zhang_delta`, `guo_vec`, `yang_omega`) with a tunable threshold where the literature value is the default.

- `hea_bench.classifiers.diagnostic_stats` — pure stdlib implementations of Wilson score intervals [13], confusion matrices, binary classifier statistics, and ROC sweeps with Youden's J [14].

- `hea_bench.benchmark` — per-source loaders (Borg, Pei, Peivaste), the cross-source consolidator, and a coverage analysis that quantifies which fraction of the consolidated dataset is scorable by the current elemental data tables.

- `hea_bench.evaluate` — the orchestrator that runs each rule against the consolidated benchmark and writes `rule_baselines.json` alongside the consolidated CSV.

Two end-user surfaces consume this library. The Python surface is a standard `pip`-installable package (`pip install hea-bench`). The browser surface is a single HTML page that loads Pyodide [15] and invokes `micropip.install()` on the same wheel that `pip` would deliver; descriptor and rule calls happen client-side in the user's browser tab, with no server-side component. *The library has one canonical implementation; both surfaces run it identically.*

2.2. Software functionalities

1. Parse arbitrary composition formulas, normalize to mole fractions, and report the six standard descriptors.
2. Apply each of the four canonical empirical rules and obtain a predicted phase classification.
3. Score a user-supplied rule (or composition set) against the consolidated v0.1.0 benchmark with full diagnostic statistics.
4. Sweep a rule threshold and report the receiver-operating-characteristic curve, optimal accuracy threshold, and optimal Youden's-J threshold.
5. Analyse element coverage of an alloy set against the descriptor data tables and report next-best-add elements.
6. Re-consolidate the per-source data via the bundled loaders and dedupe / conflict-detection logic.

2.3. The consolidated benchmark

`hea-bench` v0.1.0 ships a single canonical benchmark at `data/consolidated/v0.1.0/consolidated.csv` (7784 unique compositions). It is built by running the per-source loaders on three open datasets:

- Borg et al. [16] (740 unique `(formula, processing)` alloys; CC-BY-4.0; mirrored verbatim with SHA-256 stamp);

- Pei et al. [17] (1209 unique formulas; CC-BY-4.0; mirrored);

- Peivaste et al. [18] (7747 unique formulas after dedup of an upstream-generated combinatorial sweep; the upstream data repository declares no license and is therefore pointer-only, fetched on user request).

Compositions are merged on a 4-decimal-precision composition key. Borg's processing column is preserved as a side-channel rather than included in the join key. When two sources assign different canonical labels to the same composition, the row is flagged (`has_conflict` = 1) and the per-source labels are preserved verbatim. The v0.1.0 release contains 100 such cross-source conflicts (1.3% of the benchmark), which are themselves a useful artifact: they enumerate compositions on which the open HEA literature does not agree.

The canonical phase taxonomy is deliberately coarse for v0.1.0 — {BCC, FCC, HCP, multi-phase} — so that all three sources map into it without lossy guessing. Finer per-source labels (Borg's microstructure column with 30+ sublabels, Peivaste's 12-category column distinguishing intermetallics and amorphous alloys) are preserved as side-channel columns.

3. Illustrative examples

3.1. Cantor alloy, end to end

The canonical equimolar CoCrFeMnNi alloy serves as the field's reference HEA and as our regression sanity check. With `hea-bench`:

```
import hea_bench
cantor = hea_bench.parse_formula("CoCrFeMnNi")

hea_bench.smix(cantor)                      # 13.381 J/(mol K) = R ln 5
hea_bench.delta(cantor)                     # 3.164  %
hea_bench.vec(cantor)                       # 8.000
hea_bench.mixing_enthalpy(cantor)           # -4.16  kJ/mol
hea_bench.omega(cantor)                     # 5.794
```

All four canonical rules classify Cantor as HEA-class, single-phase, FCC — consistent with the original experimental characterization [1]. Every value above is pinned in the regression test suite.

3.2. Reference rule baselines on the consolidated benchmark

The orchestrator `python -m hea_bench.evaluate` produces the headline numbers in Table 2. These are the reference values our test suite pins; any future drift in dataset, descriptor code, vendored pair-enthalpy table, or rule thresholds also breaks the build.

The Yeh ΔS_{mix} rule is descriptive rather than predictive; 47.0% of the benchmark passes the $\Delta S_{\mathrm{mix}} > 1.5R$ HEA-class threshold, 36.9% falls in the MEA range ($R \leq \Delta S_{\mathrm{mix}} \leq 1.5R$), and 16.1% is dilute.

3.3. ROC recalibration of the Zhang δ rule

Treating the rules as proper classifiers admits a question the canonical literature does not address: *what threshold would maximize Youden's J on this benchmark?* Sweeping the Zhang δ threshold from 1 % to 12 % in 0.1 % steps yields the result in Table 3.

The canonical 6.5 % threshold is heavily skewed toward sensitivity at the expense of specificity. The balanced-optimal threshold on the consolidated benchmark is ∼2.6× tighter. Whether $J = 0.113$ constitutes a "useful" classifier is a separate question — it does not — but the recalibration finding itself is a direct, defensible output of treating the rule as a diagnostic classifier rather than a fixed cutoff.

4. Impact

`hea-bench` consolidates an open benchmark that previously did not exist in unified form, and supplies the diagnostic-classifier machinery that previously was not applied to the canonical HEA empirical rules in the open literature. We expect three classes of impact.

First, calibration of expectations. The collapse of both binary rules to near-zero Youden's J on the consolidated benchmark is a quantitative caveat for downstream users of those rules. The widely-quoted "80% accuracy" figures in the original papers reflect different, smaller datasets and a different choice of evaluation metric; on a consolidated open dataset the rules behave very differently.

Second, a substrate for future methods work. Any candidate ML model, CALPHAD-derived classifier, or new empirical descriptor can now be scored against the same dataset and the same diagnostic-stats vocabulary as the four canonical rules. The result is a head-to-head comparison rather than incommensurable single-paper accuracies.

Third, accessibility. The Pyodide front-end lets any researcher or student type a composition into a browser tab and see the same numbers the published library would produce on a local install — with the wheel itself loaded into the browser, not a re-implementation. This removes the install-friction barrier without sacrificing the library-as-canonical-implementation guarantee.

5. Conclusions

We have presented `hea-bench` 0.0.1, an open Python library and benchmark suite for high-entropy alloy phase prediction. It ships a 7784-composition consolidated benchmark drawn from three primary open sources with explicit cross-source provenance and conflict tracking; reference-baseline diagnostic- classifier evaluations of the four canonical empirical rules; a ROC-based recalibration finding for the Zhang δ rule showing a dataset-optimal threshold of $\delta < 2.5$ % rather than the canonical 6.5 %; and an in-browser Pyodide front-end that runs the same Python wheel a local `pip install` would deliver. The package is MIT-licensed, covered by 151 tests including pinned regression tests on every headline benchmark number, and intended to serve as the substrate on which subsequent ML and CALPHAD methods for HEA phase prediction can be compared.

Planned future work includes expanding the elemental-property table to lift benchmark coverage from 86.7 %

Table 2: Reference baselines for the four canonical empirical HEA phase-prediction rules, evaluated as diagnostic classifiers against the `hea-bench` v0.1.0 consolidated benchmark. n is the number of compositions in the benchmark scorable by the rule's required descriptor stack (the smaller numbers for the binary rules reflect compositions fully covered by both ELEMENTAL_DATA and the Miedema pair table; the Guo–Liu rule is stratified to single-phase observations only, as discussed in the text). Wilson 95% intervals are reported for accuracy. The two binary rules have Youden's J close to zero, indicating that they collapse to "predict single-phase almost always" on this benchmark; the Guo–Liu rule shows a strong FCC bias.

Rule	n	Accuracy	Sensitivity	Specificity
Zhang $\delta < 6.5\,\%$	6651	56.7 % [55.5 %, 57.9 %]	99.0 %	8.5 %
Yang $\Omega > 1.1$	6651	54.4 % [53.2 %, 55.6 %]	95.8 %	7.4 %
Guo–Liu VEC (stratified to FCC|BCC observed)	3463	66.9 %	92.3 % (FCC) / 48.3 % (BCC)	—

Table 3: ROC sweep of the Zhang δ rule on the `hea-bench` v0.1.0 benchmark. The canonical 6.5 % threshold yields Youden's $J = 0.075$; the dataset-optimal J is 0.113 at $\delta < 2.5\,\%$, $\sim 2.6\times$ tighter.

Threshold	Sens.	Spec.	Youden's J
6.5 % (canonical)	99.0 %	8.5 %	0.075
6.6 % (max acc.)	99.5 %	8.2 %	0.075
2.5 % (max J)	23.6 %	87.7 %	**0.113**

to $\sim$95% (the additions are low-effort because the corresponding pair-enthalpy entries are already present in the vendored Miedema table), and a v0.2.0 release that exposes the cross-source label conflicts as a structured side-dataset for downstream disagreement-resolution work.

Conflict of interest

The author declares no competing financial or non-financial interests.

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References

[1] B. Cantor, I. T. H. Chang, P. Knight, A. J. B. Vincent, Microstructural development in equiatomic multicomponent alloys, Materials Science and Engineering: A 375–377 (2004) 213–218. doi:10.1016/j.msea.2003.10.257.

[2] J.-W. Yeh, S.-K. Chen, S.-J. Lin, J.-Y. Gan, T.-S. Chin, T.-T. Shun, C.-H. Tsau, S.-Y. Chang, Nanostructured high-entropy alloys with multiple principal elements: Novel alloy design concepts and outcomes, Advanced Engineering Materials 6 (5) (2004) 299–303. doi:10.1002/adem.200300567.

[3] Y. Zhang, Y. J. Zhou, J. P. Lin, G. L. Chen, P. K. Liaw, Solid-solution phase formation rules for multi-component alloys, Advanced Engineering Materials 10 (6) (2008) 534–538. doi:10.1002/adem.200700240.

[4] S. Guo, C. T. Liu, Phase stability in high entropy alloys: Formation of solid-solution phase or amorphous phase, Progress in Natural Science: Materials International 21 (6) (2011) 433–446. doi:10.1016/S1002-0071(12)60080-X.

[5] X. Yang, Y. Zhang, Prediction of high-entropy stabilized solid-solution in multi-component alloys, Materials Chemistry and Physics 132 (2–3) (2012) 233–238. doi:10.1016/j.matchemphys.2011.11.021.

[6] A. Dunn, Q. Wang, A. Ganose, D. Dopp, A. Jain, Benchmarking materials property prediction methods: the Matbench test suite and Automatminer reference algorithm, npj Computational Materials 6 (2020) 138. doi:10.1038/s41524-020-00406-3.

[7] R. F. Zhang, S. H. Zhang, Z. J. He, J. Jing, S. H. Sheng, Miedema Calculator: A thermodynamic platform for predicting formation enthalpies of alloys within framework of Miedema's theory, Computer Physics Communications 209 (2016) 58–69. doi:10.1016/j.cpc.2016.08.013.

[8] A. R. Miedema, The electronegativity parameter for transition metals: Heat of formation and charge transfer in alloys, Journal of the Less-Common Metals 32 (1) (1973) 117–136. doi:10.1016/0022-5088(73)90078-7.

[9] A. R. Miedema, P. F. de Châtel, F. R. de Boer, Cohesion in alloys — fundamentals of a semi-empirical model, Physica B+C 100 (1) (1980) 1–28. doi:10.1016/0378-4363(80)90054-6.

[10] L. Ward, A. Dunn, A. Faghaninia, N. E. R. Zimmermann, S. Bajaj, Q. Wang, J. Montoya, J. Chen, K. Bystrom, M. Dylla, K. Chard, M. Asta, K. A. Persson, G. J. Snyder, I. Foster, A. Jain, Matminer: An open source toolkit for materials data mining, Computational Materials Science 152 (2018) 60–69. doi:10.1016/j.commatsci.2018.05.018.

[11] F. R. de Boer, R. Boom, W. C. M. Mattens, A. R. Miedema, A. K. Niessen, Cohesion in Metals: Transition Metal Alloys, Vol. 1 of Cohesion and Structure, North-Holland, Amsterdam, 1988.

[12] A. Takeuchi, A. Inoue, Classification of bulk metallic glasses by atomic size difference, heat of mixing and period of constituent elements and its application to characterization of the main alloying element, Materials Transactions 46 (12) (2005) 2817–2829. doi:10.2320/matertrans.46.2817.

[13] E. B. Wilson, Probable inference, the law of succession, and statistical inference, Journal of the American Statistical Association 22 (158) (1927) 209–212. doi:10.1080/01621459.1927.10502953.

[14] W. J. Youden, Index for rating diagnostic tests, Cancer 3 (1) (1950) 32–35. doi:10.1002/1097-0142(1950)3:1<32::AID-CNCR2820030106>3.0.CO;2-3.

[15] Pyodide contributors, Pyodide: Python with the scientific stack, compiled to WebAssembly, https://pyodide.org/, accessed 2026-05-21 (2025).

[16] C. K. H. Borg, C. Frey, J. Moy, J. Moh, D. B. Miracle, J. E. Saal, B. Meredig, T. M. Pollock, S. Gorsse, Expanded dataset of mechanical properties and observed phases of multi-principal element alloys, Scientific Data 7 (1) (2020) 430. doi:10.1038/s41597-020-00768-9.

[17] Z. Pei, J. Yin, J. A. Hawk, D. E. Alman, M. C. Gao, Machine-learning informed prediction of high-entropy solid solution formation: Beyond the Hume-Rothery rules, npj Computational Materials 6 (1) (2020) 50. doi:10.1038/s41524-020-0308-7.

[18] I. Peivaste, E. Jossou, A. A. Tiamiyu, Data-driven analysis and prediction of stable phases for high-entropy alloy design, Scientific Reports 13 (1) (2023) 22556. doi:10.1038/s41598-023-50044-0.